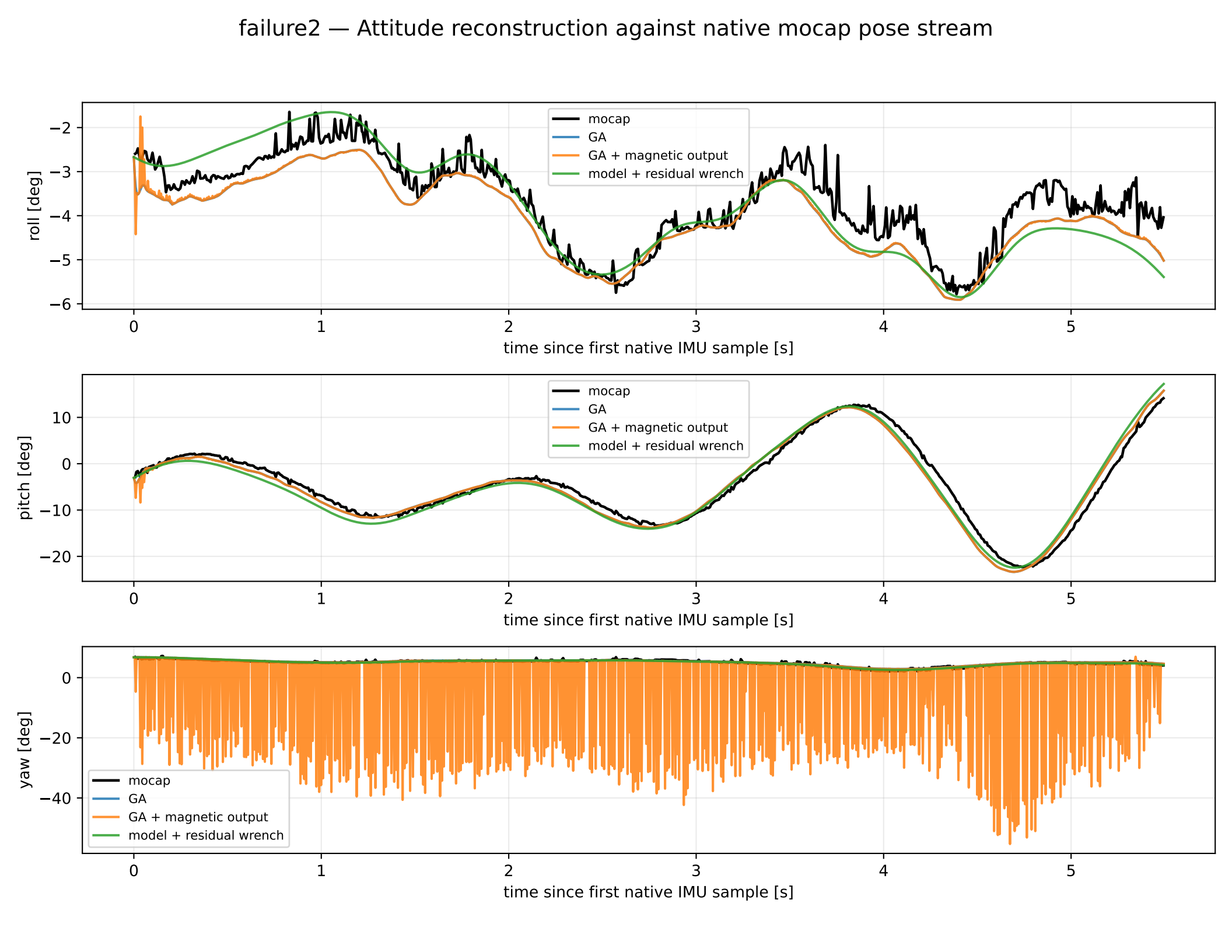
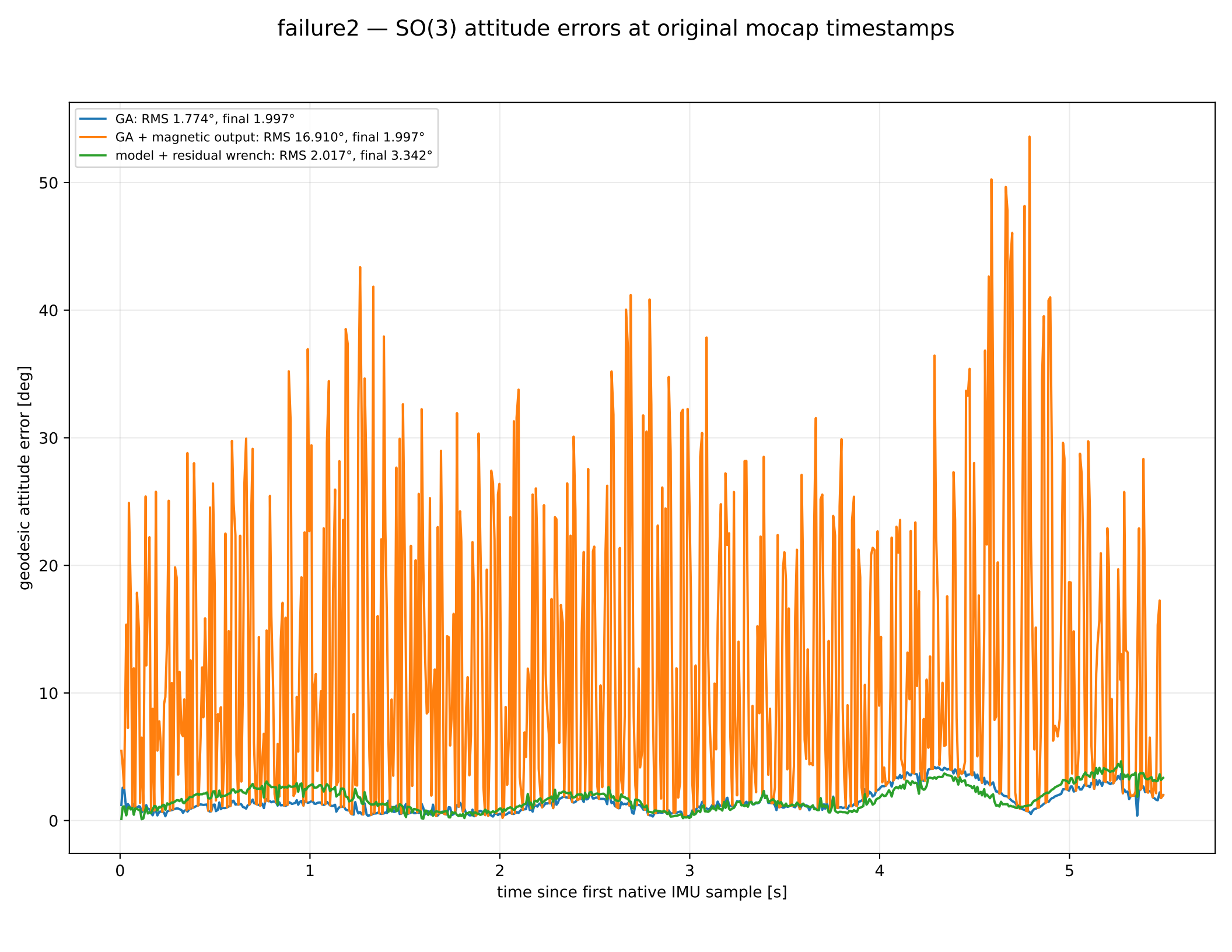
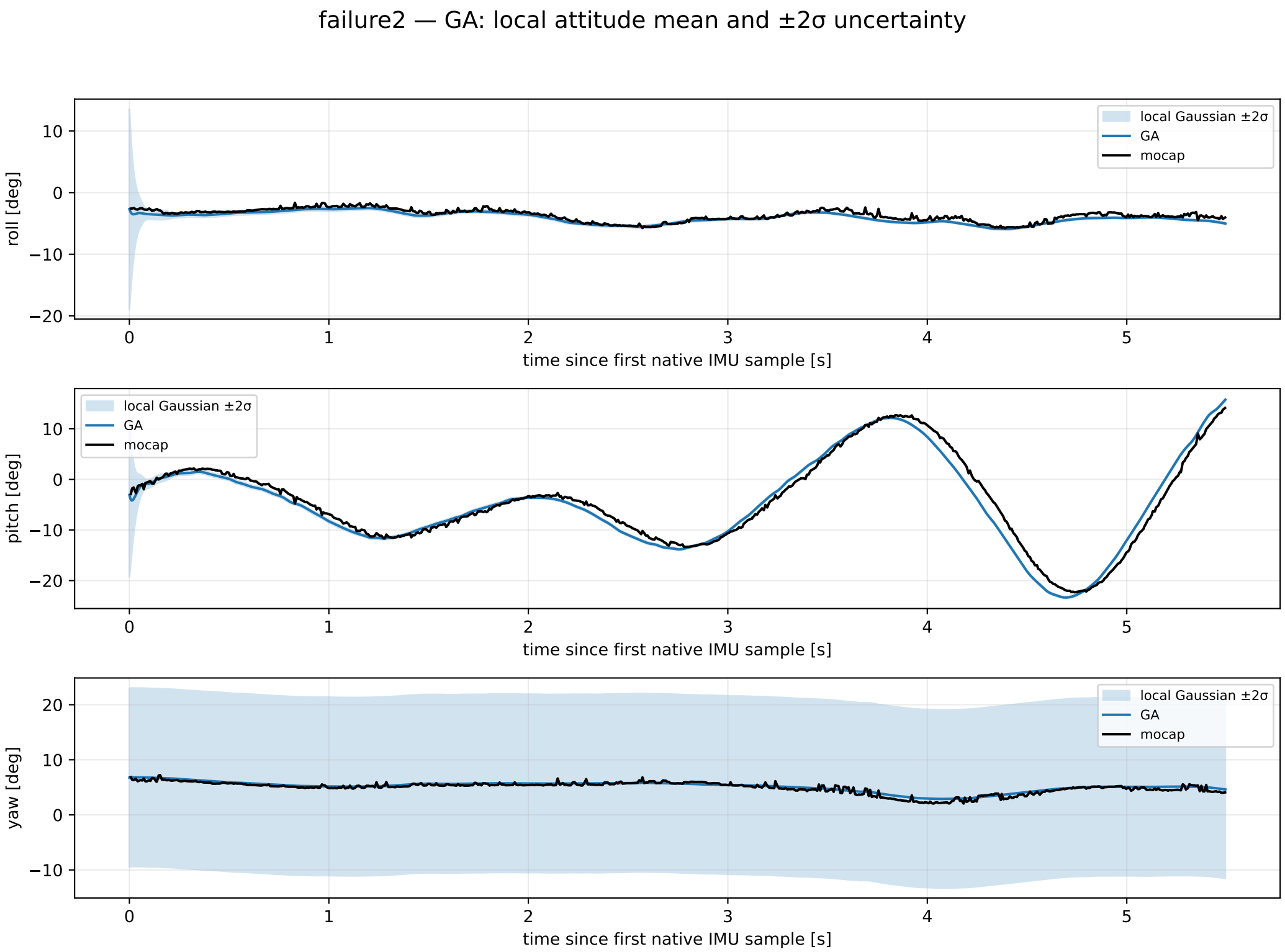
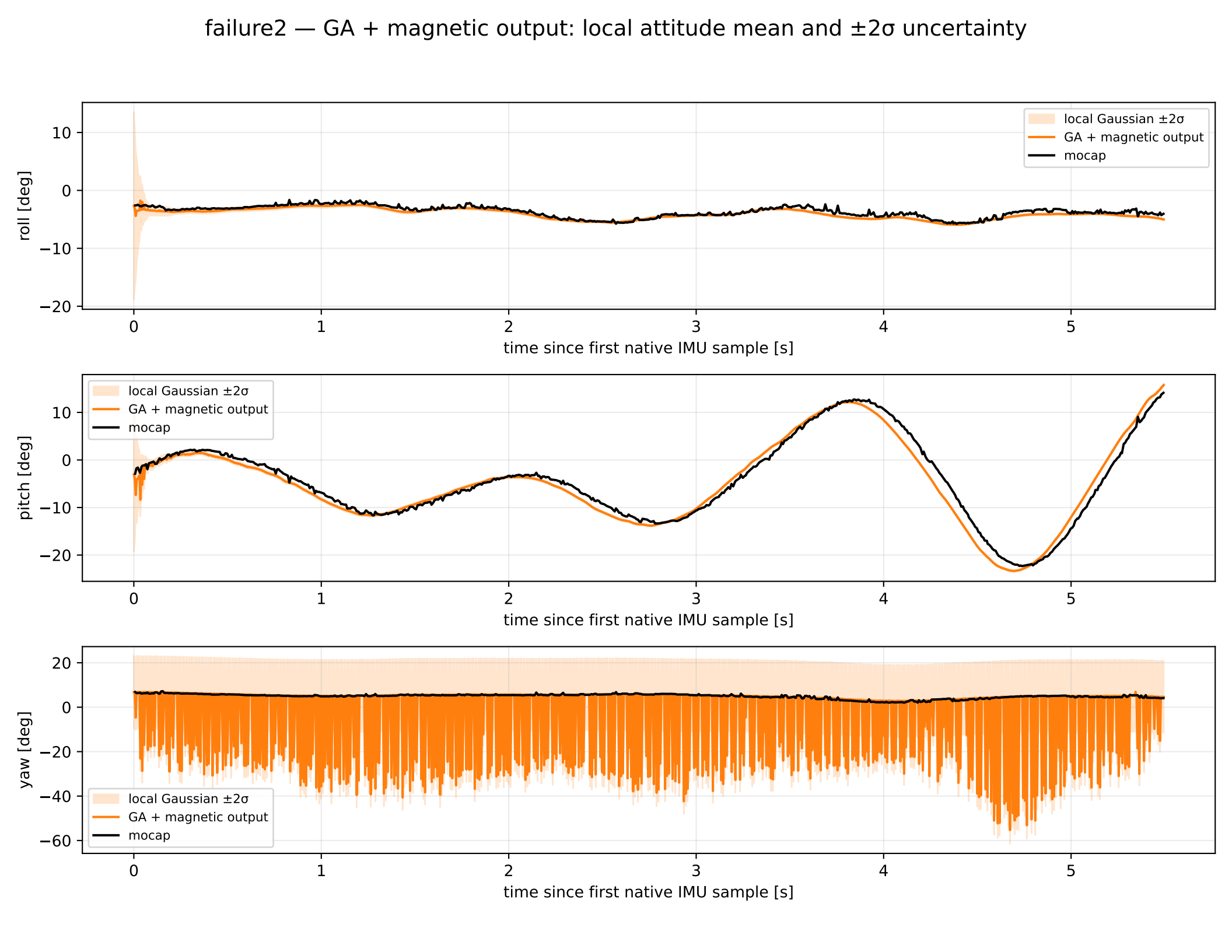
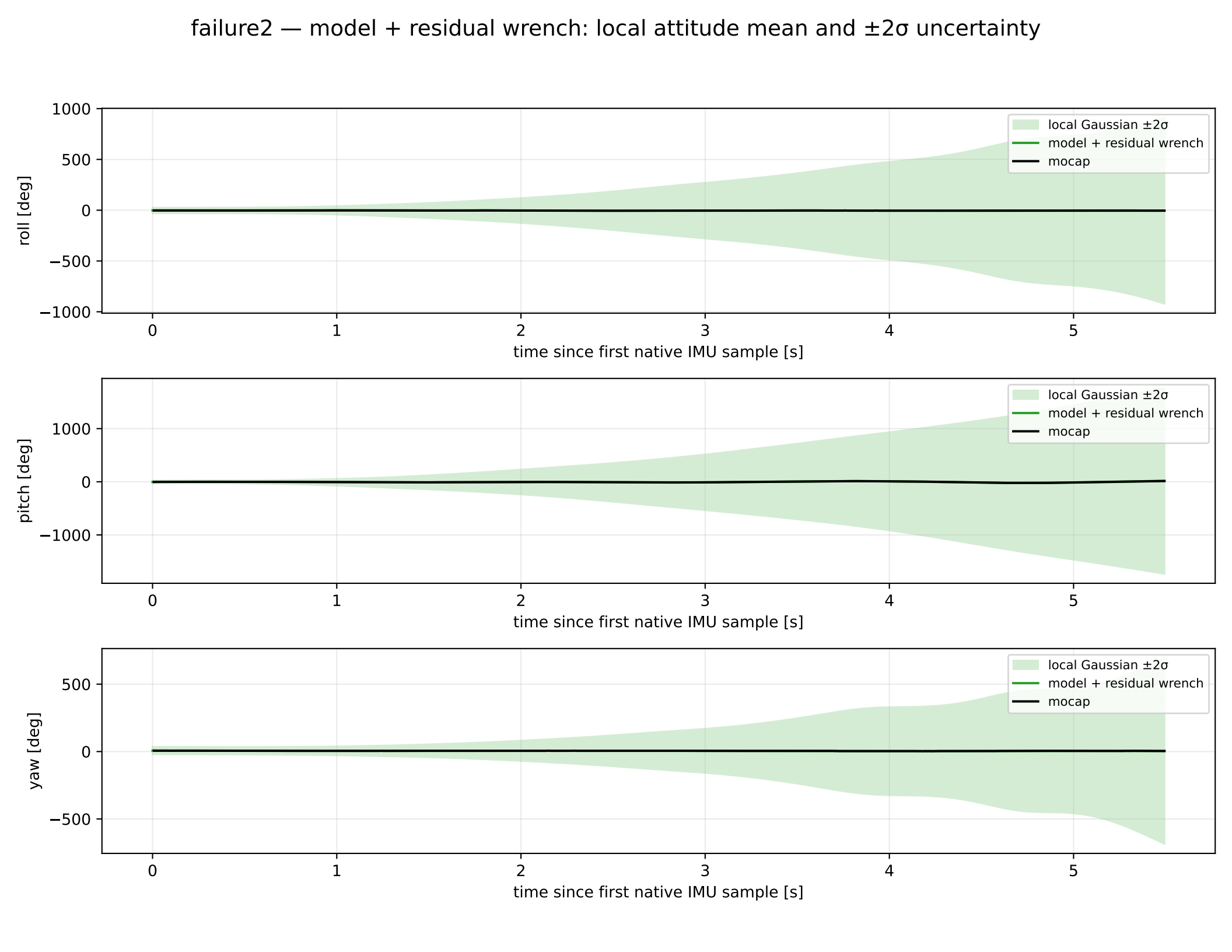

failure2 — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

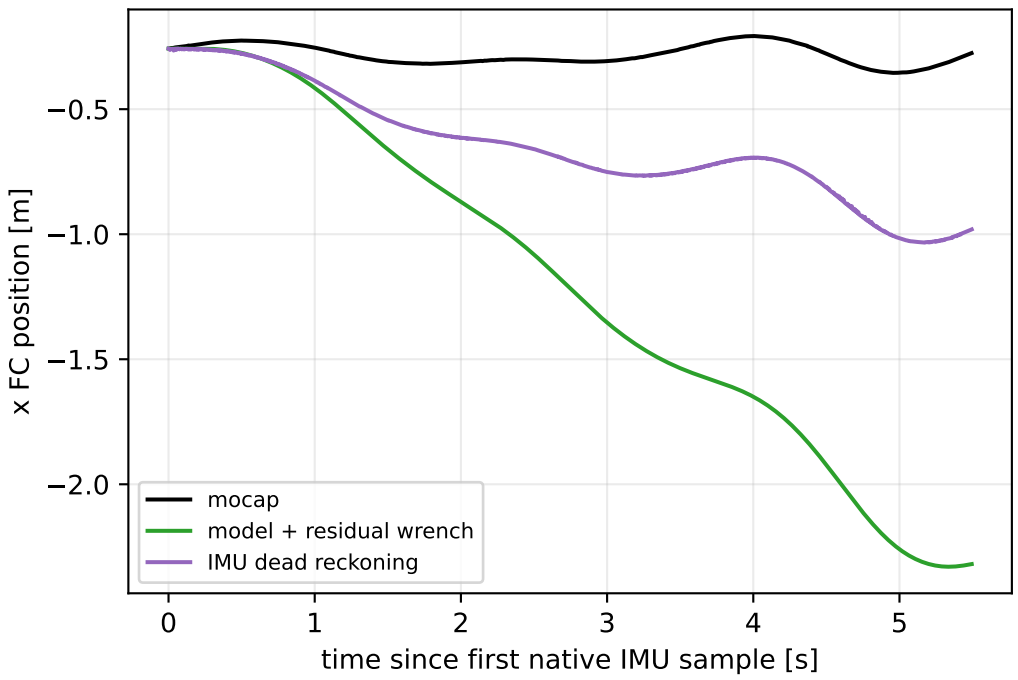

failure2 — SO(3) attitude errors at original mocap timestamps
GA: RMS 1.774°, final 1.997°
GA + magnetic output: RMS 16.910°, final 1.997°
model + residual wrench: RMS 2.017°, final 3.342°
geodesic attitude error [deg]
time since first native IMU sample [s]
0
10
20
30
40
50
0
1
2
3
4
5

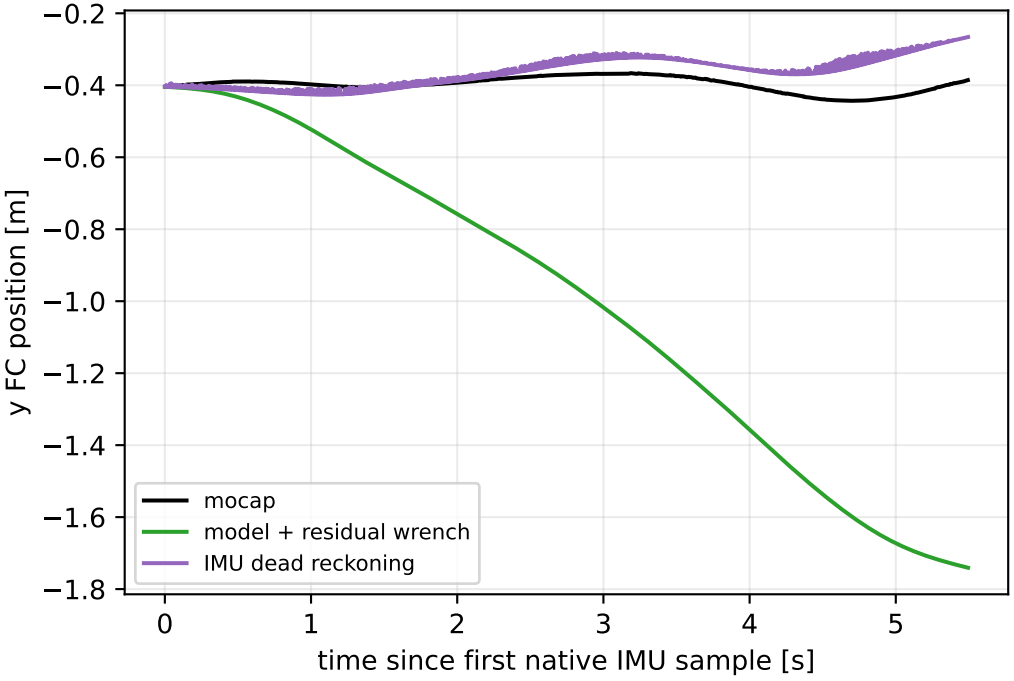

failure2 — GA: local attitude mean and ±2σ uncertainty
local Gaussian ±2σ
GA
mocap
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]

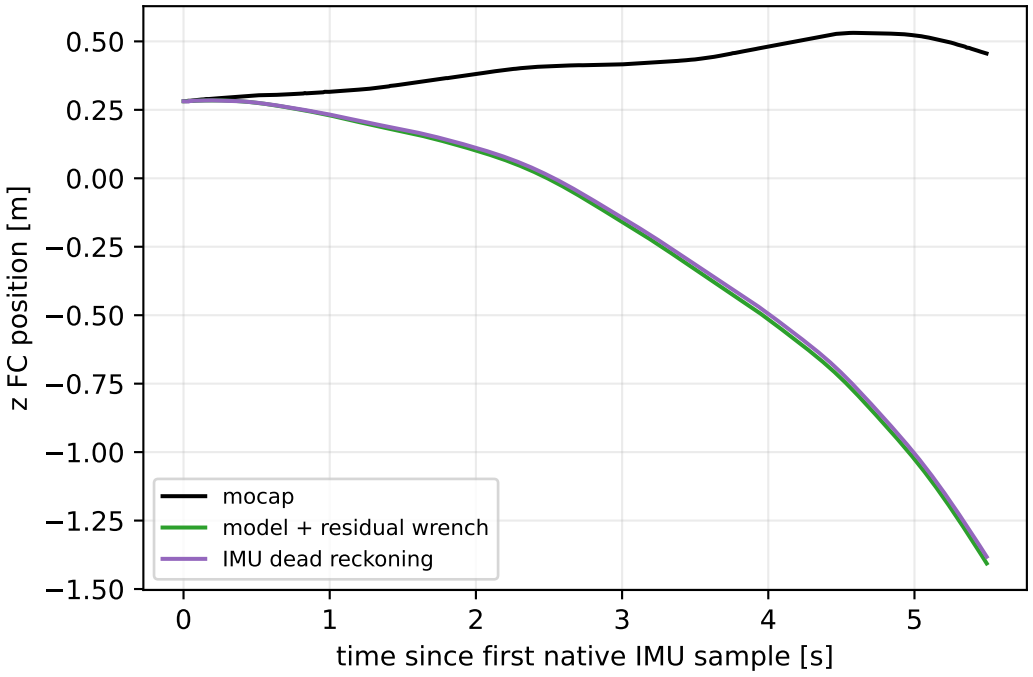

failure2 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

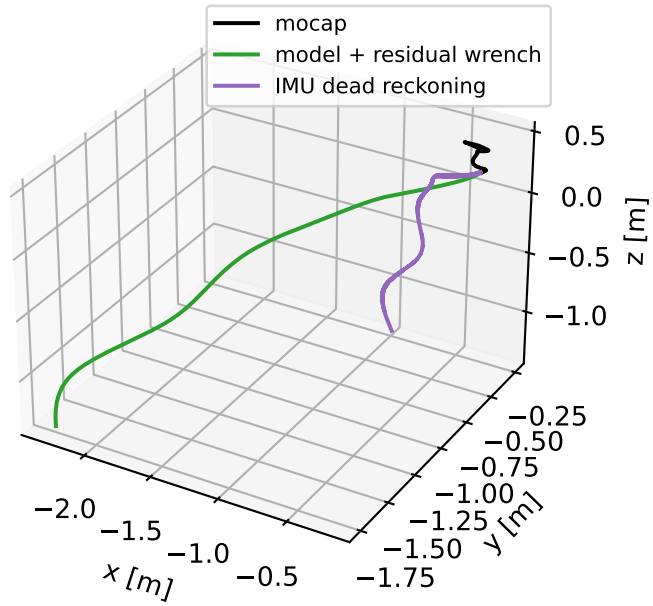

failure2 — model + residual wrench: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
model + residual wrench
mocap

failure2 — Position reconstruction (mean trajectories)



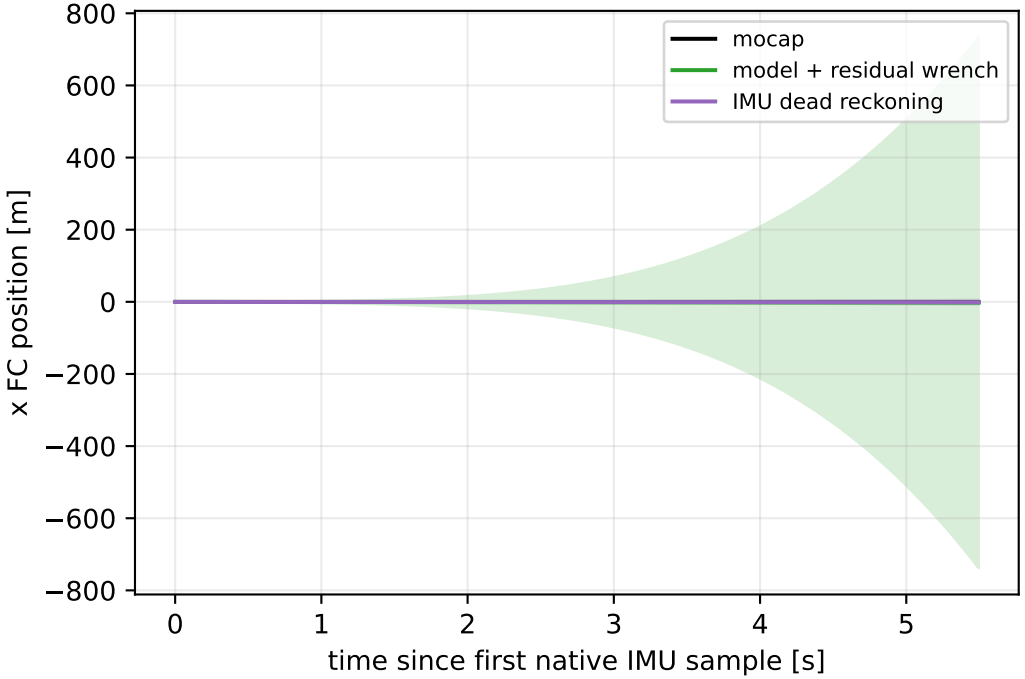


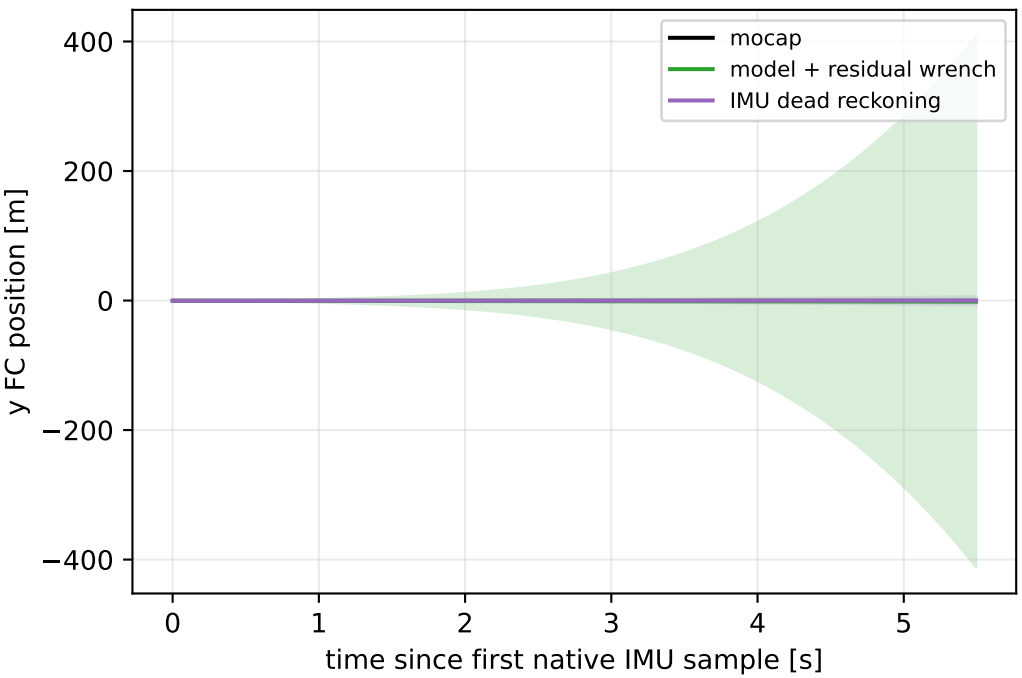


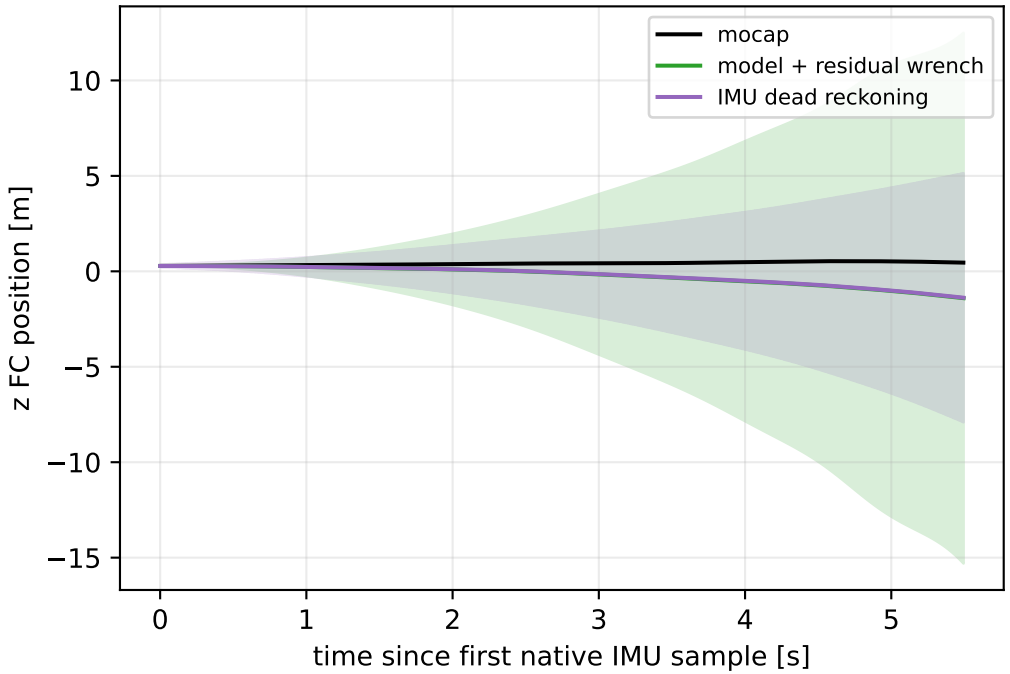


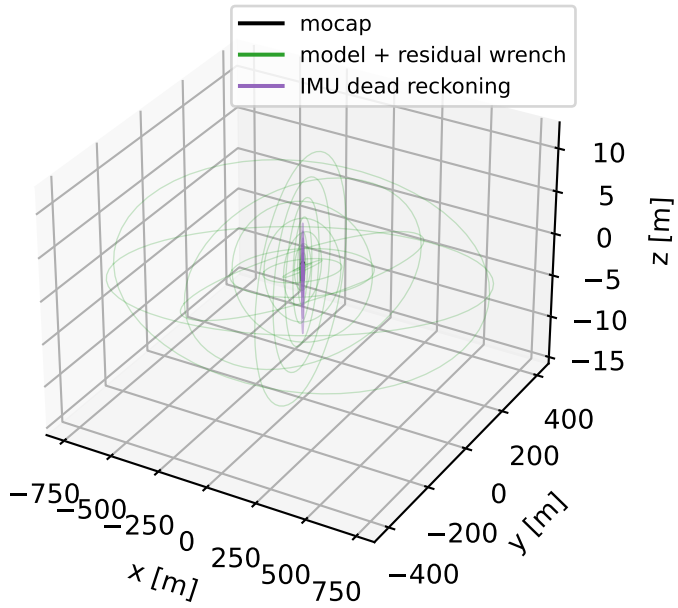
Residual-wrench model replay and IMU integration are not reset to later mocap.

failure2 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.